

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250285249

Kind Code

A1

Publication Date

September 11, 2025

Inventor(s)

Yuan; Le et al.

Image Processing Method and Apparatus and Terminal Device

Abstract

An image processing method includes determining a format of first image data; determining a first interface mode that is currently used, where the first interface mode is determined when the processing device establishes physical connections with a display device, and the format of the first image data format is not supported by the first interface mode; and processing the first image data based on the first interface mode to generate second image data, where a format of the second image data is supported by the first interface mode. Image data will be processed when an image format of the image data is not supported by the first interface mode.

Inventors: Yuan; Le (Shenzhen, CN), Wang; Zheng (Shenzhen, CN), Yan; Xiumei (Shenzhen, CN)

Applicant: Huawei Technologies Co., Ltd. (Shenzhen, CN)

Family ID: 1000008618159

Appl. No.: 19/018393

Filed: January 13, 2025

Foreign Application Priority Data

CN 201710266941.3

Apr. 21, 2017

Related U.S. Application Data

parent US continuation 17581193 20220121 parent-grant-document US 12217404 child US 19018393

parent US continuation 16596269 20191008 parent-grant-document US 11263730 child US 17581193

parent WO continuation PCT/CN2017/114565 20171205 PENDING child US 16596269

Publication Classification

Int. Cl.: G06T5/92 (20240101); G06T11/00 (20060101)

U.S. Cl.:

CPC G06T5/92 (20240101); G06T11/001 (20130101); G06T2207/10016 (20130101); G06T2207/20208 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 17/581,193, filed on Jan. 21, 2022, which is a continuation of U.S. patent application Ser. No. 16/596,269, filed on Oct. 8, 2019, now U.S. Pat. No. 11,263,730, which is a continuation of International Application No. PCT/CN2017/114565, filed on Dec. 5, 2017. The International Application claims priority to Chinese Patent Application No. 201710266941.3, filed on Apr. 21, 2017. All of the afore-mentioned patent applications are hereby incorporated by reference in their entireties.

TECHNICAL FIELD

[0002] This application relates to the field of display processing technologies, and more specifically, to an image processing method and apparatus and terminal device in the field of display processing technologies.

BACKGROUND

[0003] Compared with a conventional standard dynamic range (SDR) video, a high dynamic range (HDR) video, as a new video technology, may greatly improve viewing experience of an audience.

[0004] For a hybrid video stream that includes both an HDR video (for example, a movie) and an SDR video (for example, an advertisement), when the hybrid video stream arrives at a playback device, the playback device sends a high definition multimedia interface (HDMI) reset signal to a display device (for example, the display device supports HDR video playing). The reset signal is used to force the display device to reinitialize an HDMI standard. Therefore, when receiving the hybrid video stream, the display device can sense a video mode change of the hybrid video stream in terms of time, convert the hybrid video stream into a play mode supported by the display device, and finally play a converted video.

[0005] In the foregoing solution, when detecting that a video stream includes both the HDR video and the SDR video, the playback device sends the HDMI reset signal to the display device. However, when switching between the HDR video and the SDR video occurs for a plurality of times in a segment of the video stream, the playback device needs to send the HDMI reset signal to the display device for a plurality of times.

[0006] Because the display device reinitializes the HDMI standard for a relatively long time, frequently sending the HDMI reset signal to the display device by the playback device causes one or more seconds of black screen of the display device, severely affecting viewing experience of a user.

SUMMARY

[0007] This application provides an image processing method and apparatus, to avoid one or more seconds of black screen of a display device, thereby improving viewing experience of a user.

[0008] According to a first aspect, an image processing method is provided, and is applied to a display system that supports N types of interface modes, where $N \geq 1$, and the display system includes a processing device and a display device. The method includes determining, by the

processing device in the display system, a format of first image data, where the format of the first image data is a first image data format, determining, by the processing device, a first interface mode that is currently used, where the first interface mode is determined when the processing device establishes physical connections with the display device, and the first image data format is not an image format supported by the first interface mode, and processing, by the processing device, the first image data based on the first interface mode to generate second image data, where a format of the second image data is a second image data format, and the second image data format is the image format supported by the first interface mode.

[0009] Image data will be processed when an image format of the image data is not supported by the first interface mode (the first interface mode is determined when the processing device establishes physical connections with the display device) such that a format of processed image data can be supported by the first interface mode (namely, the format of the second image data), to avoid one or more seconds of black screen of the display device resulting from re-initialization of an interface mode between the processing device and the display device caused by inconsistency of the format of the first image data and the first interface mode, thereby improving viewing experience of a user.

[0010] With reference to the first aspect, in a first implementation of the first aspect, the method further includes sending, by the processing device, the second image data to the display device, where the processing device is a playback device.

[0011] The processing device sends the processed image data to the display device such that the display device can display the processed image data (namely, the second image data) in a uniform format in order to avoid one or more seconds of black screen of the display device caused by inconsistency between a format of image data and an image format supported by the first interface mode, thereby improving viewing experience of a user.

[0012] With reference to the first aspect or the foregoing implementation, in a second implementation of the first aspect, the processing, by the processing device, the first image data to generate second image data includes converting, by the processing device, the first image data from a first color space to a second color space, performing, by the processing device, dynamic range mapping on a luminance component of image data corresponding to the second color space, where the image data corresponding to the second color space is obtained after color space conversion is performed on the first image data, restoring, by the processing device, image data obtained after performing the dynamic range mapping on the luminance component into a linear optical space, performing, by the processing device, a color gamut mapping on image data restored into the linear optical space, and performing, by the processing device, non-uniform encoding on image data obtained after performing the color gamut mapping, to generate the second image data.

[0013] With reference to the first aspect and the foregoing implementations, in a third implementation of the first aspect, the first image data format is an image format corresponding to an SDR, and the second image data format is an image format corresponding to an HDR, or the first image data format is an image format corresponding to an HDR, and the second image data format is an image format corresponding to an SDR, or the first image data format is an image format corresponding to an HDR of a hybrid log-gamma (HLG) standard, and the second image data format is an image format corresponding to an HDR of a perceptual quantizer (PQ) standard, or the first image data format is an image format corresponding to an HDR of a PQ standard, and the second image data format is an image format corresponding to an HDR of an HLG standard.

[0014] With reference to any one of the first aspect and the foregoing implementations, in a fourth implementation of the first aspect, the first interface mode is an HDMI operation mode.

[0015] With reference to any one of the first aspect and the foregoing implementations, in a fifth implementation of the first aspect, the first image data includes a video image and/or a graphic image.

[0016] Image processing is performed on the video image, image processing is also performed on a

graphic image related to the video image, and a processed video image and a processed graphic image are sent to the display device together, thereby improving viewing experience about the graphic image of a user.

[0017] According to a second aspect, an image processing apparatus is provided, and is applied to a display system that supports N types of interface modes, where $N \geq 1$, and the display system includes the apparatus and a display device. The apparatus includes a determining unit configured to determine a format of first image data, where the format of the first image data is a first image data format, the determining unit is further configured to determine a first interface mode that is currently used, the first interface mode is determined when the apparatus establishes physical connections with the display device, and the first image data format is not an image format supported by the first interface mode, and a processing unit configured to process the first image data based on the first interface mode to generate second image data, where a format of the second image data is a second image data format, and the second image data format is an image format supported by the first interface mode.

[0018] Image data will be processed when an image format of the image data is not supported by the first interface mode (the first interface mode is determined when the apparatus establishes physical connections with the display device) such that a format of processed image data (namely, the format of the second image data) can be supported by the first interface mode, to avoid one or more seconds of black screen of the display device resulting from reinitialization of an interface mode between the apparatus and the display device caused by inconsistency of the format of the first image data and the first interface mode, thereby improving viewing experience of a user.

[0019] With reference to the second aspect, in a first implementation of the second aspect, the apparatus further includes a sending unit configured to send the second image data to the display device, where the apparatus is a playback device.

[0020] The apparatus sends the processed image data to the display device such that the display device can display the processed image data (namely, the second image data) in a uniform format, to avoid one or more seconds of black screen of the display device caused by inconsistency between a format of image data and an image format supported by the first interface mode, thereby improving viewing experience of a user.

[0021] With reference to the second aspect or the foregoing implementation, in a second implementation of the second aspect, the processing unit is configured to convert, the first image data from a first color space to a second color space, perform, dynamic range mapping on a luminance component of image data corresponding to the second color space, where the image data corresponding to the second color space is obtained after color space conversion is performed on the first image data, restore image data obtained after performing the dynamic range mapping on the luminance component into a linear optical space, perform color gamut mapping on image data restored into the linear optical space, and perform non-uniform encoding on image data obtained after performing the color gamut mapping, to generate the second image data.

[0022] With reference to the second aspect and the foregoing implementations, in a third implementation of the second aspect, the first image data format is an image format corresponding to an SDR, and the second image data format is an image format corresponding to an HDR, or the first image data format is an image format corresponding to an HDR, and the second image data format is an image format corresponding to an SDR, or the first image data format is an image format corresponding to an HDR of an HLG standard, and the second image data format is an image format corresponding to an HDR of a PQ standard, or the first image data format is an image format corresponding to an HDR of a PQ standard, and the second image data format is an image format corresponding to an HDR of an HLG standard.

[0023] With reference to any one of the second aspect and the foregoing implementations, in a fourth implementation of the second aspect, the first interface mode is an HDMI operation mode.

[0024] With reference to any one of the second aspect and the foregoing implementations, in a fifth

implementation of the second aspect, the first image data includes a video image and/or a graphic image.

[0025] Image processing is performed on the video image, image processing is also performed on a graphic image related to the video image, and a processed video image and a processed graphic image are sent to the display device together, thereby improving viewing experience about the graphic image of a user.

[0026] According to a third aspect, an image processing apparatus is provided, and is applied to a display system that supports N types of interface modes, where $N \geq 1$, and the display system includes the image processing apparatus and a display device. The image processing apparatus includes a memory and a processor. The memory is configured to store instructions. The processor is configured to invoke the instructions stored in the memory to perform the following steps: determining a format of first image data, where the format of the first image data is a first image data format, and determining a first interface mode that is currently used, where the first interface mode is determined when the image processing apparatus establishes physical connections with the display device, and the first image data format is not an image format supported by the first interface mode. The processor is further configured to process the first image data based on the first interface mode to generate second image data, where a format of the second image data is a second image data format, and the second image data format is the image format supported by the first interface mode.

[0027] Image data will be processed when an image format of the image data is not supported by the first interface mode (the first interface mode is determined when the image processing apparatus establishes physical connections with the display device) such that a format of processed image data can be supported by the first interface mode (namely, the format of the second image data), to avoid one or more seconds of black screen of the display device resulting from re-initialization of an interface mode between the image processing apparatus and the display device caused by inconsistency of the format of the first image data and the first interface mode, thereby improving viewing experience of a user.

[0028] With reference to the third aspect, in a first implementation of the third aspect, the image processing apparatus further includes an output interface configured to send the second image data to the display device, where the image processing apparatus is a playback device.

[0029] The image processing apparatus sends the processed image data to the display device such that the display device can display the processed image data (namely, the second image data) in a uniform format, to avoid one or more seconds of black screen of the display device caused by inconsistency between a format of image data and an image format supported by the first interface mode, thereby improving viewing experience of a user.

[0030] With reference to the third aspect or the foregoing implementation, in a second implementation of the third aspect, the processor is configured to convert the first image data from a first color space to a second color space, perform dynamic range mapping on a luminance component of image data corresponding to the second color space, where the image data corresponding to the second color space is obtained after color space conversion is performed on the first image data, restore image data obtained after performing the dynamic range mapping on the luminance component into a linear optical space, perform color gamut mapping on image data restored into the linear optical space, and perform non-uniform encoding on image data obtained after performing the color gamut mapping, to generate the second image data.

[0031] With reference to the third aspect and the foregoing implementations, in a third implementation of the third aspect, the first image data format is an image format corresponding to an SDR, and the second image data format is an image format corresponding to an HDR, or the first image data format is an image format corresponding to an HDR, and the second image data format is an image format corresponding to an SDR, or the first image data format is an image format corresponding to an HDR of an HLG standard, and the second image data format is an

image format corresponding to an HDR of a PQ standard, or the first image data format is an image format corresponding to an HDR of a PQ standard, and the second image data format is an image format corresponding to an HDR of an HLG standard.

[0032] With reference to any one of the third aspect and the foregoing implementations, in a fourth implementation of the third aspect, the first interface mode is an HDMI operation mode.

[0033] With reference to any one of the third aspect and the foregoing implementations, in a fifth implementation of the third aspect, the first image data includes a video image and/or a graphic image.

[0034] Image processing is performed on the video image, image processing is also performed on a graphic image related to the video image, and a processed video image and a processed graphic image are sent to the display device together, thereby improving viewing experience about the graphic image of a user.

[0035] According to a fourth aspect, a computer storage medium is provided, where the computer storage medium stores program code, and the program code is used to indicate operations for implementing the image processing method in the first aspect or any optional implementations of the first aspect.

Description

DESCRIPTION OF DRAWINGS

[0036] FIG. 1 is a schematic structural diagram of a processing device for image processing according to this application.

[0037] FIG. 2 is a schematic flowchart of an image processing method according to this application.

[0038] FIG. 3 is another schematic flowchart of an image processing method according to this application.

[0039] FIG. 4 is a schematic block diagram of an image processing apparatus according to this application.

[0040] FIG. 5 is a schematic structural diagram of an image processing apparatus according to this application.

DESCRIPTION OF EMBODIMENTS

[0041] The following describes the technical solutions of this application with reference to the accompanying drawings.

[0042] FIG. 1 is a schematic structural diagram of a processing device in a display system to which this application is applicable. The processing device includes components such as an input interface **110**, a video decoder **120**, a central processing unit (CPU) **130**, a video image processing module **140**, an output interface **150**, and an internal memory **160**.

[0043] A person skilled in the art may understand that the structural diagram of the processing device shown in FIG. 1 is merely an example instead of a limitation. The processing device may further include more or fewer components than those shown in the figure, or may include different component settings.

[0044] First, a video data stream is transmitted from the input interface **110** (for example, a wired network port or a wireless network port) to the processing device, and is stored in the internal memory **160** using an internal data bus. The CPU **130** and the video decoder **120** decode the video data stream stored in the internal memory **160**. The video image processing module **140** accesses the internal memory **160** using the internal data bus, obtains decoded video data from the internal memory **160**, determines a format of the video based on the decoded video data, and further determines whether to perform format conversion on the video data.

[0045] Then, the video image processing module **140** outputs video data (for example, video image

data or graphic image data) obtained after format conversion or unconverted video data (for example, video image data or graphic image data) to a display device (for example, a television set) through the output interface **150**.

[0046] It should be noted that, usually, the CPU **130** may directly access the video decoder **120** and the video image processing module **140**. This is not limited in this application.

[0047] FIG. **2** is a schematic flowchart of an image display method **200** according to this application. As shown in FIG. **2**, the method **200** is applied to a display system that supports N types of interface modes, where $N \geq 1$, the display system includes a processing device and a display device, and the method **200** includes the following steps.

[0048] **S210**. The processing device in the display system determines a format of first image data, where the format of the first image data is a first image data format.

[0049] **S220**. The processing device determines a first interface mode that is currently used, where the first interface mode is determined when the processing device establishes physical connections with the display device, and the first image data format is not an image format supported by the first interface mode.

[0050] **S230**. The processing device processes the first image data based on the first interface mode to generate second image data, where a format of the second image data is a second image data format, and the second image data format is the image format supported by the first interface mode.

[0051] In an embodiment, the processing device in the display system first determines the format of the received first image data, namely, the first image data format. The display system in which the processing device is located supports N types of interface modes. After determining the format of the first image data that is currently received, the processing device further determines an interface mode that is currently used (namely, the first interface mode). The first interface mode is determined through negotiation when the processing device establishes physical connections with the display device. The first image data format is not an image format supported by the first interface mode, in an embodiment, the first interface mode does not support transmission of the first image data between the processing device and the display device.

[0052] Therefore, the processing device needs to convert the format of the first image data based on the first interface mode such that the first interface mode can support the transmission of the first image data between the processing device and the display device.

[0053] For example, the processing device processes the first image data to generate the second image data. The format of the second image data is an image format supported by the first interface mode, in an embodiment, the first interface mode can support transmission of the second image data between the processing device and the display device.

[0054] It should be noted that the first image data and the second image data include video images and/or graphic images. In an embodiment, the first image data may be a video image or a graphic image, or the first image data may include both the video image and graphic image. The graphic image may include a control menu (for example, a volume adjustment progress bar or a video fast forward/rewind adjustment progress bar that is displayed on a display device), or the graphic image may be image data (for example, a static image shown in a video) included in video data. This is not limited in this application.

[0055] As an example instead of a limitation, the processing device may be a playback device (for example, a set top box). In addition, after performing format conversion on the first image data, the processing device sends the second image data to the display device, where the second image data is obtained after the format conversion of the first image data. The second image data is displayed by the display device.

[0056] It should be understood that the foregoing merely uses an example in which the processing device is a playback device for descriptions. However, this application is not limited thereto. The processing device may be another device (for example, a mobile phone device). In addition, the device has a function of obtaining a video data stream, decoding the video data stream, and sending

the video data stream to the display device. This is not limited in this application.

[0057] As an example instead of a limitation, the processing, by the processing device, the first image data to generate the second image data includes the following steps. In step 1, the processing device converts the first image data from a first color space to a second color space, in step 2, the processing device performs dynamic range mapping on a luminance component of image data corresponding to the second color space, where the image data corresponding to the second color space is obtained after color space conversion is performed on the first image data, in step 3, the processing device restores image data obtained after the dynamic range mapping on the luminance component into a linear optical space, in step 4, the processing device performs color gamut mapping on image data obtained after the linear optical space restoration, and in step 5, the processing device performs non-uniform encoding on image data obtained after the color gamut mapping, to generate the second image data.

[0058] In an embodiment, after determining that decoded video image data (namely, the first image data) needs format conversion, the processing device first performs color space conversion on a video signal corresponding to the first image data, in an embodiment, converts the video signal from the first color space (for example, a luminance, chroma: blue, chroma: red (YCbCr) color space) to the second color space (for example, an red, green, blue (RGB) color space), where the RGB color space is suitable for adjusting an image range. After converting the video signal from the YCbCr color space to the RGB color space, the processing device performs dynamic range mapping on a luminance component corresponding to the video signal obtained after the color space conversion, in an embodiment, maps a dynamic range of the luminance component of the video signal to a dynamic range of a luminance component conforming to a video signal output format. The processing device performs restoration on a video signal conforming to the output format into the linear optical space, and then performs the color gamut mapping on the video signal after the restoration. During the color gamut mapping, the video signal in the linear optical space is mapped from a source color gamut to an output color gamut. The processing device performs non-uniform encoding on the video signal (namely, the video signal mapped to the output color gamut) to generate the second image data such that the format of the second image data conforms to an output format requirement of an output video signal.

[0059] Therefore, a processing procedure of converting the format of the first image data into the format of the second image data is completed.

[0060] It should be noted that execution sequences of the steps 1 to 5 shall not constitute any limitation on this application. For example, step 1, step 3, and step 4 may be performed before step 2 and step 5, which is not limited in this application.

[0061] As an example instead of a limitation, the first image data format is an image format corresponding to an SDR, and the second image data format is an image format corresponding to an HDR, or the first image data format is an image format corresponding to an HDR, and the second image data format is an image format corresponding to an SDR, or the first image data format is an image format corresponding to an HDR of an HLG standard, and the second image data format is an image format corresponding to an HDR of a PQ standard, or the first image data format is an image format corresponding to an HDR of a PQ standard, and the second image data format is an image format corresponding to an HDR of an HLG standard.

[0062] It should be noted that the foregoing method for processing the first image data is applicable to conversion between any two foregoing different image formats.

[0063] It should be understood that, in this application, conversion between the foregoing image formats is only used as an example to describe the image processing method in this application. However, this application is not limited thereto. The technical solutions of this application may further support format conversion between other different image formats, and this is not limited in this application.

[0064] As an example instead of a limitation, the first interface mode is an HDMI operation mode.

[0065] According to the image processing method in this application, image data corresponding to an image format that is not supported by the first interface mode (the first interface mode is determined when the processing device establishes physical connections with the display device) will be processed such that a format of processed image data can be supported by the first interface mode (namely, the format of the second image data), to avoid one or more seconds of black screen of the display device resulting from re-initialization of an interface mode between the processing device and the display device caused by inconsistency of the format of the first image data and the first interface mode, thereby improving viewing experience of a user.

[0066] FIG. 3 is another schematic flowchart of an image processing method **300** according to this application.

[0067] The following describes the method **300** using an example in which a first image data format is an image format corresponding to an SDR, and a first interface mode is an interface mode that supports an image format corresponding to an HDR. The method **300** includes the following steps.

[0068] **S301**. Receive video data.

[0069] **S302**. Decode the video data to determine an image format corresponding to the video data.

[0070] In an embodiment, a processing device first receives video data transmitted from an input interface **110**, and then decodes the video data to determine the image format corresponding to the video data. For example, the processing device determines that the image format corresponding to the video data is the image format corresponding to the SDR.

[0071] **S303**. Determine whether the image format corresponding to the video data is the image format corresponding to the HDR.

[0072] In an embodiment, after determining the image format corresponding to the input video data, the processing device needs to determine whether the image format corresponding to the video data is consistent with an image format supported by the first interface mode, and the first interface mode here supports the image format corresponding to the HDR. In other words, the processing device needs to determine whether the image format corresponding to the video data is the image format corresponding to the HDR.

[0073] It should be noted that the first interface mode may be determined when the processing device establishes physical connections with the display device, in other words, the image format supported by the first interface mode is determined when the processing device establishes physical connections with the display device.

[0074] **S304**. Convert a format of a video image corresponding to the video data into the image format corresponding to the HDR.

[0075] **S305**. Convert a format of a graphic image corresponding to the video data into the image format corresponding to the HDR.

[0076] In an embodiment, when determining that the image format corresponding to the video data is not the image format corresponding to the HDR, the processing device performs format conversion on the video image and the graphic image that correspond to the video data, in an embodiment, converts the formats of the video image and the graphic image that correspond to the video data into the image format corresponding to the HDR.

[0077] **S306**. Output video data corresponding to the HDR.

[0078] In an embodiment, when **S303** is performed, if it is determined that the image format corresponding to the video data is the image format corresponding to the HDR, **S306** (in an embodiment, the video data is output) is directly performed after **S303**. Alternatively, when **S303** is performed, if it is determined that the image format corresponding to the video data is not the image format corresponding to the HDR, **S304** and **S305** (in an embodiment, by performing **S304** and **S305**, the formats of the images corresponding to the video data are converted) are continuously performed after **S303**, and **S306** is performed after **S304** and **S305**. Finally, the format of the first image data is converted from the image format corresponding to the SDR into the image

format corresponding to the HDR (namely, the format of the second image data).

[0079] It should be noted that execution sequences of the processes in the foregoing method **300** shall not constitute any limitation on the technical solutions of this application, in other words, the execution sequences of the processes in the method **300** are not limited in this application.

[0080] The image processing method according to this application is described above in detail with reference to FIG. 1 to FIG. 3. The following describes an image processing apparatus according to this application in detail with reference to FIG. 4.

[0081] FIG. 4 is a schematic block diagram of an image processing apparatus **400** according to this application. The apparatus **400** is applied to a display system that supports N types of interface modes, where $N \geq 1$, and the display system includes the apparatus and a display device. As shown in FIG. 4, the apparatus **400** includes a determining unit **410** and a processing unit **420**.

[0082] The determining unit **410** is configured to determine a format of first image data, where the format of the first image data is a first image data format.

[0083] The determining unit **410** is further configured to determine a first interface mode that is currently used, where the first interface mode is determined when the apparatus establishes physical connections with the display device, and the first image data format is not an image format supported by the first interface mode.

[0084] The processing unit **420** is configured to process the first image data based on the first interface mode to generate second image data, where a format of the second image data is a second image data format, and the second image data format is the image format supported by the first interface mode.

[0085] Optionally, the apparatus **400** further includes a sending unit **430** configured to send the second image data to the display device, where the apparatus is a playback device.

[0086] Optionally, the processing unit **420** is configured to convert the first image data from a first color space to a second color space, perform dynamic range mapping on a luminance component of image data corresponding to the second color space, where the image data corresponding to the second color space is obtained after color space conversion is performed on the first image data, restore image data obtained after performing the dynamic range mapping on the luminance component into a linear optical space, perform color gamut mapping on image data restored into the linear optical space, and perform non-uniform encoding on image data obtained after performing the color gamut mapping, to generate the second image data.

[0087] Optionally, the first image data format is an image format corresponding to an SDR, and the second image data format is an image format corresponding to an HDR, or the first image data format is an image format corresponding to an HDR, and the second image data format is an image format corresponding to an SDR, or the first image data format is an image format corresponding to an HDR of an HLG standard, and the second image data format is an image format corresponding to an HDR of a PQ standard, or the first image data format is an image format corresponding to an HDR of a PQ standard, and the second image data format is an image format corresponding to an HDR of an HLG standard. Optionally, the first interface mode is an HDMI operation mode.

[0088] Optionally, the first image data includes a video image and/or a graphic image.

[0089] The image processing apparatus **400** according to this application may be corresponding to an implementation body of the image processing method **200** according to this application. In addition, units of the image processing apparatus **400** and other operations and/or functions described above are used to implement corresponding procedures of the method **200** in FIG. 2 respectively. For brevity, details are not described herein again.

[0090] According to the image processing apparatus in this application, image data corresponding to an image format that is not supported by the first interface mode (the first interface mode is determined when the image processing apparatus establishes physical connections with the display device) is processed such that a format of processed image data (namely, the format of the second

image data) can be supported by the first interface mode, to avoid one or more seconds of black screen of the display device resulting from re-initialization of an interface mode between the image processing apparatus and the display device caused by inconsistency of the format of the first image data and the first interface mode, thereby improving viewing experience of a user.

[0091] The image processing method according to this application is described above in detail with reference to FIG. 1 to FIG. 3. The following describes an image processing apparatus according to this application in detail with reference to FIG. 5.

[0092] FIG. 5 is a schematic block diagram of an image processing apparatus **500** according to this application. The image processing apparatus **500** is applied to a display system that supports N types of interface modes, where $N \geq 1$, and the display system includes the image processing apparatus and a display device. As shown in FIG. 5, the image processing apparatus **500** includes a processor **510** and a memory **520**. The memory **520** is configured to store instructions, and the processor **510** is configured to execute the instructions stored in the memory **520**.

[0093] The memory **520** may include a volatile memory, for example, a random access memory (RAM). The memory may alternatively include a nonvolatile memory, for example, a flash memory, a hard disk drive (HDD), or a solid-state drive (SSD). The memory **520** may alternatively include a combination of the memories of the foregoing types.

[0094] The processor **510** may be a central processing unit (CPU), a network processor (NP), or a combination of the CPU and the NP. The processor **510** may further include a hardware chip. The hardware chip may be an application-specific integrated circuit (ASIC), a programmable logic device (PLD), or a combination thereof. The PLD may be a complex programmable logic device (CPLD), a field-programmable gate array (FPGA), a generic array logic (GAL), or any combination thereof.

[0095] The processor **510** is configured to determine a format of the first image data, where the format of the first image data is a first image data format.

[0096] The processor **510** is configured to determine a first interface mode that is currently used, where the first interface mode is determined when the image processing apparatus establishes physical connections with the display device, and the first image data format is not an image format supported by the first interface mode.

[0097] The processor **510** is further configured to process the first image data based on the first interface mode to generate second image data, where a format of the second image data is an image format supported by the first interface mode.

[0098] Optionally, the image processing apparatus **500** further includes an output interface **530** configured to send the second image data to the display device, where the image processing apparatus is a playback device.

[0099] Optionally, the processor **510** is configured to convert the first image data from a first color space to a second color space, perform dynamic range mapping on a luminance component of image data corresponding to the second color space, where the image data corresponding to the second color space is obtained after color space conversion is performed on the first image data, restore image data obtained after performing the dynamic range mapping on the luminance component into a linear optical space, perform color gamut mapping on image data restored into the linear optical space, and perform non-uniform encoding on image data obtained after performing the color gamut mapping, to generate the second image data.

[0100] Optionally, the first image data format is an image format corresponding to an SDR, and the second image data format is an image format corresponding to an HDR, or the first image data format is an image format corresponding to an HDR, and the second image data format is an image format corresponding to an SDR, or the first image data format is an image format corresponding to an HDR of an HLG standard, and the second image data format is an image format corresponding to an HDR of a PQ standard, or the first image data format is an image format corresponding to an HDR of a PQ standard, and the second image data format is an image format

corresponding to an HDR of an HLG standard.

[0101] Optionally, the first interface mode is an HDMI operation mode.

[0102] Optionally, the first image data includes a video image and/or a graphic image.

[0103] The image processing apparatus **500** according to this application may be corresponding to an implementation body of the image processing method **200** according to this application. In addition, units of the image processing apparatus **500** and other operations and/or functions described above are used to implement corresponding procedures of the method **200** in FIG. 2 respectively. For brevity, details are not described herein again.

[0104] According to the image processing apparatus in this application, image data will be processed when an image format of the image data is not supported by the first interface mode (the first interface mode is determined when the image processing apparatus establishes physical connections with the display device) such that a format of processed image data (namely, the format of the second image data) can be supported by the first interface mode, to avoid one or more seconds of black screen of the display device resulting from reinitialization of an interface mode between the image processing apparatus and the display device caused by inconsistency of the format of the first image data and the first interface mode, thereby improving viewing experience of a user.

[0105] It should be understood that sequence numbers of the foregoing processes do not mean execution sequences in the embodiments of this application. The execution sequences of the processes should be determined based on functions and internal logic of the processes, and shall not constitute any limitation on the implementation processes of this application.

[0106] A person of ordinary skill in the art may be aware that, in combination with the examples described in the embodiments disclosed in this specification, units and algorithm steps may be implemented by electronic hardware or a combination of computer software and electronic hardware. Whether the functions are executed by hardware or software depends on particular applications and design constraint conditions of the technical solutions. A person skilled in the art may use different methods to implement the described functions for each particular application, but it should not be considered that the implementation goes beyond the scope of this application.

[0107] It may be clearly understood by a person skilled in the art that, for convenient and brief description, for a specific working process of the foregoing system, apparatus, and unit, refer to a support process in the foregoing method embodiments, and details are not described herein again.

[0108] In the embodiments provided in this application, it should be understood that the disclosed system, apparatus, and method may be implemented in another manner. For example, the described apparatus embodiments are merely examples. For example, the unit division is merely logical function division and may be another division in an embodiment. For example, a plurality of units or components may be combined or integrated into another system, or some features may be ignored or not performed. In addition, the displayed or discussed mutual couplings or direct couplings or communication connections may be implemented through some interfaces. Indirect couplings or communication connections between the apparatuses or units may be implemented in electronic, mechanical, or other forms.

[0109] The units described as separate parts may or may not be physically separate, and parts displayed as units may or may not be physical units, may be located in one position, or may be distributed on a plurality of network units. Some or all of the units may be selected based on requirements in an embodiment to achieve the objectives of the solutions of the embodiments.

[0110] In addition, functional units in the embodiments of this application may be integrated into one processing unit, or each of the units may exist alone physically, or two or more units are integrated into one unit.

[0111] When the functions are implemented in a form of a software functional unit and sold or used as an independent product, the functions may be stored in a computer-readable storage medium. Based on such an understanding, the technical solutions of this application essentially, or a part

contributing to other approaches, or some of the technical solutions may be implemented in a form of a software product. The computer software product is stored in a storage medium, and includes several instructions for instructing a computer device (which may be a personal computer, a server, a network device, or the like) to perform all or some of the steps of the methods described in the embodiments of this application. The foregoing storage medium includes any medium that can store program code, such as a USB flash drive, a removable hard disk, a read-only memory (ROM), a RAM, a magnetic disk, or an optical disc.

[0112] The foregoing descriptions are merely specific implementations of this application, but are not intended to limit the protection scope of this application. Any variation or replacement readily figured out by a person skilled in the art within the technical scope disclosed in this application shall fall within the protection scope of this application. Therefore, the protection scope of this application shall be subject to the protection scope of the claims.

Claims

1. An image processing method implemented by a processing device in a display system, wherein the image processing method comprises: determining a first image data format associated with first image data; determining a first interface mode being used between the processing device and a display device of the display system; making a determination that the first image data format is incompatible with the first interface mode; and processing, in response to the determination, the first image data to generate second image data in a second image data format that is supported by the first interface mode.
2. The image processing method of claim 1, further comprising sending the second image data to the display device, wherein the processing device is a playback device.
3. The image processing method of claim 1, wherein processing the first image data comprises: converting the first image data from a first color space to a second color space to obtain converted image data corresponding to the second color space; performing dynamic range mapping on a luminance component of the converted image data to obtain mapped data; restoring the mapped data into a linear optical space to obtain restored data; performing color gamut mapping on the restored data to obtain gamut mapped data; and performing non-uniform encoding on the gamut mapped data to generate the second image data.
4. The image processing method of claim 3, wherein the first image data format is standard dynamic range (SDR) and the second image data format is high dynamic range (HDR), the first image data format is HDR and the second image data format is SDR, the first image data format is HDR of a hybrid log-gamma (HLG) standard and the second image data format is HDR of a perceptual quantizer (PQ) standard, or the first image data format is HDR of a PQ standard and the second image data format is HDR of an HLG standard.
5. The image processing method of claim 1, wherein the first interface mode is a High-Definition Multimedia Interface (HDMI) operation mode.
6. The image processing method of claim 1, wherein the first image data comprises a video image or a graphic image.
7. An image processing apparatus, comprising: a memory configured to store one or more programs; and a processor coupled to the memory and configured to execute the one or more programs to cause the image processing apparatus to: determine a first image data format associated with first image data; determine a first interface mode being used between the image processing apparatus and a display device; make a determination that the first image data format is incompatible with the first interface mode; and process, in response to the determination, the first image data to generate second image data in a second image data format that is supported by the first interface mode.
8. The image processing apparatus of claim 7, wherein the image processing apparatus further

comprises an output interface configured to send the second image data to the display device, and wherein the image processing apparatus is a playback device.

9. The image processing apparatus of claim 7, wherein the processor is further configured to execute the one or more programs to cause the image processing apparatus to: convert the first image data from a first color space to a second color space to obtain converted image data corresponding to the second color space; perform dynamic range mapping on a luminance component of the converted image data to obtain mapped data; restore the mapped data into a linear optical space to obtain restored data; perform color gamut mapping on the restored data to obtain gamut mapped data; and perform non-uniform encoding on the gamut mapped data to generate the second image data.

10. The image processing apparatus of claim 9, wherein the first image data format is standard dynamic range (SDR) and the second image data format is high dynamic range (HDR), the first image data format is HDR and the second image data format is SDR, the first image data format is HDR of a hybrid log-gamma (HLG) standard and the second image data format is HDR of a perceptual quantizer (PQ) standard, or the first image data format is HDR of a PQ standard and the second image data format is HDR of an HLG standard.

11. The image processing apparatus of claim 7, wherein the first interface mode is a High-Definition Multimedia Interface (HDMI) operation mode.

12. The image processing apparatus of claim 7, wherein the first image data comprises a video image or a graphic image.

13. A computer program product comprising computer-executable instructions that are stored on a non-transitory computer-readable medium and that, when executed by a processor, cause an image processing apparatus to: determine a first image data format associated with first image data; determine a first interface mode being used between the image processing apparatus and a display device; make a determination that the first image data format is incompatible with the first interface mode; and process, in response to the determination, the first image data to generate a second image data in a second image data format that is supported by the first interface mode.

14. The computer program product of claim 13, wherein when executed by the processor, the computer-executable instructions further cause the image processing apparatus to send the second image data to the display device.

15. The computer program product of claim 13, wherein when executed by the processor, the computer-executable instructions further cause the image processing apparatus to: convert the first image data from a first color space to a second color space to obtain converted image data corresponding to the second color space; perform dynamic range mapping on a luminance component of the converted image data to obtain mapped data; restore the mapped data into a linear optical space to obtain restored data; perform color gamut mapping on the restored data to obtain gamut mapped data; and perform non-uniform encoding on the gamut mapped data to generate the second image data.

16. The computer program product of claim 15, wherein the first image data format is standard dynamic range (SDR) and the second image data format is high dynamic range (HDR), or the first image data format is HDR and the second image data format is SDR.

17. The computer program product of claim 13, wherein the first interface mode is a High-Definition Multimedia Interface (HDMI) operation mode.

18. The computer program product of claim 13, wherein the first image data comprises a video image or a graphic image.

19. The computer program product of claim 15, wherein the first image data format is HDR of a hybrid log-gamma (HLG) standard and the second image data format is HDR of a perceptual quantizer (PQ) standard.

20. The computer program product of claim 15, wherein the first image data format is HDR of a

perceptual quantizer (PQ) standard and the second image data format is HDR of a hybrid log-gamma (HLG) standard.
